

JAMESTOWN 38 WSW, ND, USA

WMO: 727510

Lat: 46.7702N

Lon: 99.4778W

Elev: 585

StdP: 94.49

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 54937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.4	-26.2	-31.1	0.2	-28.0	-28.8	0.3	-25.8	13.9	-9.1	12.8	-9.4	2.5	N/A	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	30.9	21.3	29.1	20.2	27.5	19.4	22.9	28.3	21.7	27.2	20.6	26.0	4.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.1	26.0	20.0	15.8	24.9	18.8	14.6	23.6	71.1	28.1	66.1	27.1	62.3	26.0	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	9.9	8.7	-32.8	35.2	2.5	2.2	-34.6	36.8	-36.1	38.0	-37.5	39.3	-39.3	40.9		
(5)			-32.4	25.3	2.4	1.6	-34.1	26.4	-35.5	27.3	-36.8	28.2	-38.5	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-12.3	-11.7	-3.5	4.1	11.7	17.3	20.5	19.3	14.7	6.2	-1.8	-10.3	(6)
(7)		DBStd	13.15	7.87	7.27	7.89	6.00	4.99	3.57	3.14	3.32	4.51	5.29	6.64	7.34	(7)
(8)		HDD10.0	3086	691	607	422	189	40	2	0	0	11	142	354	629	(8)
(9)		HDD18.3	5198	949	840	677	427	211	58	14	27	127	378	603	887	(9)
(10)		CDD10.0	1117	0	0	3	12	94	221	324	289	150	24	1	0	(10)
(11)		CDD18.3	189	0	0	0	0	6	28	80	58	16	1	0	0	(11)
(12)		CDH23.3	1853	0	0	1	5	81	249	742	573	189	14	0	0	(12)
(13)		CDH26.7	525	0	0	0	1	13	48	222	190	49	3	0	0	(13)
(14)	Wind	WSAvg	3.7	3.9	3.8	4.0	4.5	4.1	3.4	2.8	3.3	4.0	3.7	3.6	(14)	
(15)	Precipitation	PrecAvg	510	12	10	19	29	74	88	91	54	53	43	16	13	(15)
(16)		PrecMax	701	30	34	68	55	162	183	256	128	101	109	57	30	(16)
(17)		PrecMin	385	1	1	2	7	7	45	19	11	5	3	0	2	(17)
(18)		PrecStd	83	9	9	17	14	50	34	53	32	27	30	15	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	6.5	19.9	23.4	28.6	30.1	33.1	34.2	31.5	25.1	17.5	7.3	(19)
(20)			MCWB	2.1	2.8	10.5	12.5	16.9	21.0	22.6	20.7	20.3	15.5	10.2	3.6	(20)
(21)		2%	DB	2.7	3.4	15.8	19.3	25.5	28.1	31.0	30.9	27.8	20.7	13.0	3.5	(21)
(22)			MCWB	0.6	1.2	9.8	10.6	15.8	19.4	21.9	20.6	18.1	12.5	7.9	1.1	(22)
(23)		5%	DB	0.9	1.2	11.2	16.6	23.3	26.1	29.1	28.7	25.4	17.5	9.6	1.3	(23)
(24)			MCWB	-0.5	-0.2	6.4	8.8	14.9	18.3	21.1	20.1	16.9	11.0	5.4	-0.3	(24)
(25)		10%	DB	-0.9	-0.9	6.5	13.9	20.8	24.5	27.5	26.5	23.0	14.6	6.8	-0.7	(25)
(26)			MCWB	-1.9	-2.0	3.7	7.3	13.5	17.6	20.3	18.8	16.0	9.7	3.8	-1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.8	13.7	13.3	19.0	22.9	25.9	23.5	21.7	16.2	10.6	4.0	(27)
(28)			MCDWB	6.8	6.6	16.9	20.9	25.0	28.1	29.7	29.6	28.2	23.5	17.3	7.0	(28)
(29)		2%	WB	0.9	1.7	9.9	10.9	17.8	21.1	23.7	21.9	20.1	14.0	7.7	1.7	(29)
(30)			MCDWB	2.7	3.4	17.0	18.1	24.1	25.9	28.3	28.1	25.3	19.8	12.5	3.8	(30)
(31)		5%	WB	-0.1	0.3	6.8	9.2	16.2	19.9	22.4	21.0	18.5	11.8	5.3	0.1	(31)
(32)			MCDWB	1.3	1.7	11.8	15.7	20.7	25.1	27.4	26.5	23.7	17.0	8.2	1.6	(32)
(33)		10%	WB	-1.5	-1.5	4.2	7.7	14.7	18.6	21.2	20.0	17.0	10.1	3.6	-1.4	(33)
(34)			MCDWB	-0.4	-0.3	7.5	13.0	19.3	23.5	26.7	25.2	21.6	14.6	6.2	-0.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	9.4	9.6	11.0	11.9	11.2	11.8	12.3	12.2	10.5	9.9	8.4	(35)
(36)			MCDBR	10.9	10.7	15.7	16.3	15.0	13.6	13.7	14.9	15.5	15.5	13.9	10.4	(36)
(37)			MCWBR	9.4	8.7	9.5	9.0	7.4	7.2	6.4	6.6	7.5	8.5	8.7	7.9	(37)
(38)		5% WB	MCDBR	10.7	10.8	15.5	15.9	12.2	12.1	11.8	13.0	13.7	14.4	13.1	9.7	(38)
(39)			MCWBR	9.2	8.6	9.6	8.9	6.4	7.0	6.9	6.4	7.3	8.4	8.5	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.300	0.298	0.358	0.370	0.368	0.374	0.375	0.345	0.296	0.267	0.252	(40)	
(41)		taud	2.256	2.302	2.441	2.296	2.340	2.400	2.397	2.376	2.433	2.551	2.509	2.455	(41)	
(42)		Ebn at Noon	781	868	934	896	895	896	885	870	868	876	840	822	(42)	
(43)		Edh at Noon	80	94	95	122	121	115	114	111	96	73	61	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.69	2.81	3.93	4.96	5.67	6.30	6.53	5.48	4.16	2.61	1.65	1.25	(44)	
(45)		RadStd	0.15	0.29	0.31	0.39	0.43	0.42	0.32	0.43	0.27	0.33	0.21	0.11	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page